Causal assumptions in triangulation

Complete methodological discussion draft

All eight website pages, including expanded reference material.

Internal links and the contents below navigate within this PDF. Editable templates are available from the website downloads.

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Making causal assumptions explicit in triangulation

DRAFT FOR FEEDBACK

The problem

Cross-stream agreement in triangulation is currently interpreted without making explicit the causal structure, and applicability assumptions that connect each stream to the target question. Shared biases can manufacture apparent agreement.

The conceptual claim to stress-test

Target relationship → Stream-specific contribution → Connecting assumptions

This is the reasoning I want to stress-test: whether it captures what warrants a cross-stream judgement, including when shared bias could explain agreement. The methodological aim is to make that reasoning explicit and reviewable; the templates are its practical application.

Where this work sits, methodologically

Directed acyclic graphs (DAGs) are familiar tools for making the assumptions behind **confounding control within a study** explicit. This proposal extends that logic to the **between-stream setting**: which target relationship each evidence stream informs, and which identifying, proxy, and applicability assumptions connect its contribution to that target. An **estimand layer** underpins this: each question and each evidence stream gets a specified target quantity at protocol stage (Component A), and each stream's quantity is checked for causal alignment with the review target at evidence-summary stage (Component B).

What I mean by the framework

These working notes call this **"scaffolding"**: a linked set of visual and written tools that helps a team make its assumptions explicit and carry its reasoning between stages of a living systematic review.

The DAG visualises the hypothesised causal structure. An **assumption register** records the assumptions, their rationale, and what could challenge them. The **Summary of Evidence table (SET)** connects the findings to those assumptions. The three templates guide how the team prepares, discusses, and reports this information.

See the worked TAAR1 DAG and register →

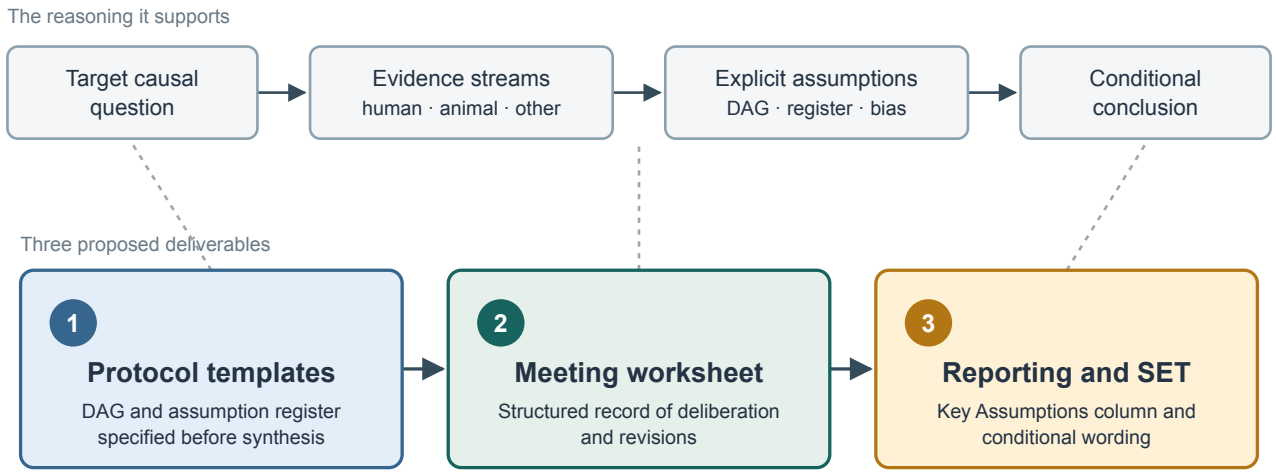


Figure 1: Three proposed deliverables — protocol templates, a meeting worksheet, and reporting with a SET — and the reasoning each supports.

Three deliverables as methods contributions

The 3 templates are the practical application of the framework.

Proposal	What would change?	Practical output
1. Protocol templates	Use a DAG to identify assumptions and plan how evidence could inform them.	Proposed wording for both protocol routes.
2. Triangulation meeting worksheet	Record how evidence and bias affect interpretation of the DAG and its assumptions.	A four-step worksheet to complete in Word.
3. Reporting and evidence summary	Connect the conclusion to the assumptions it depends on.	Existing SET columns plus Key Assumptions, and reporting wording.

Two applications, at different stages

Application	Stage and purpose
TAAR1	A retrospective reconstruction showing what the framework could make explicit in a completed triangulation.
LSR8 · Heat & Mental Health	An ongoing application, introduced after protocol completion while the living systematic reviews are being conducted.

Feedback sought

1. **Conceptual:** Is the proposed reasoning appropriate: a target relationship or causal query, the stream-specific evidence contribution, and the assumptions connecting them?
2. **Conceptual:** Does the assumption register clarify the distinction between causal structure, identification, proxy validity, and applicability? What is missing or redundant?
3. **Practical:** What is the minimum useful information for a protocol, a triangulation meeting, and a published SET? Who should maintain it across updates?
4. **Evaluation:** What observations in LSR8 would show whether the framework changes the clarity, consistency, or burden of deliberation?
5. **Estimand layer:** Does specifying the target quantity per question and per stream (and checking alignment at the evidence summary) clarify the workflow, or add burden without changing the decisions?

This discussion concerns qualitative appraisal and documentation. Quantitative parameterisation and weighting are later developments. About and references.

Concepts: the terms needed for this proposal

The argument

The proposed unit of reasoning is **target relationship → stream-specific contribution → connecting assumptions**. A cross-stream judgement needs to specify what each stream informs and the conditions under which its findings bear on the target question. Shared bias remains a possible explanation for apparent agreement.

The framework extends familiar within-study DAG reasoning to this between-stream setting. The DAG shows the proposed causal structure; the register records the connecting assumptions; the Summary of Evidence table (SET) links the findings to them. Its role as a qualitative counterpart to quantitative bias modelling is explained in the Briefing.

A worked illustration

The TAAR1 example shows the DAG and its assumption register together. Selected labels identify the proxy connection and the proposed mechanism. The worked evidence summary shows how the register links enter the SET.

Ten terms for this discussion

Term	Working definition
Causal DAG — directed acyclic graph	A diagram of hypothesised direct causal influences, with no directed cycles. Both arrows and omitted arrows encode assumptions.
Assumption register	A versioned record of assumptions with IDs, statements, affected relationships/streams, judgements, rationale, and what could challenge them.
Proxy	A measured variable used to stand in for another construct. Its relevance needs justification; the animal behavioural outcome and human symptom outcome remain distinct.
Estimand	The precisely specified quantity a question seeks: a descriptive quantity, an association parameter (marginal or adjusted), a causal contrast, or — for hypothesis-generating questions — a candidate relationship that may have no single numerical estimand. An estimand can be specified even when the available data cannot identify it.
Causal alignment assessment	A structured judgement of whether the quantity a stream estimates, or the contribution it makes, addresses the review's target causal question, a related question, or a different link in the causal pathway — and what assumptions are needed to connect it to the target. Distinct from indirectness (applicability of a quantity) and from risk of bias (validity of a quantity for its own estimand). Assessed narratively at the evidence-summary stage; not scored.
Integration level	The level at which evidence streams can legitimately be integrated, determined by the causal alignment assessment: **A — sufficiently common causal effect** (quantitative integration may be possible); **B — related but non-identical effects or constructs** (retain within-stream synthesis; compare across streams without pooling); **C — different links in a broader causal explanation** (complementary, pathway-/question-level triangulation).
Stream-level estimand	The quantity an evidence stream can estimate in its own terms: population, intervention/exposure, comparator, outcome, timing, summary quantity and effect type. Specified before alignment is judged, so that alignment relates two specified quantities — for example, an animal

Term	Working definition
	behavioural contrast and a human clinical contrast — rather than a study and an underspecified ambition.
Indirectness / applicability	A difference between the evidence and the target population, exposure/intervention, outcome, timing, or setting that affects its relevance. A within-stream judgement: a difference between the evidence a stream contains and the population, exposure, outcome, timing, or setting specified in that stream's own estimand. Judged in the SET's indirectness column, one rating per stream row. Whether the stream's quantity matches the review's target is an alignment judgement.
SET — Summary of Evidence table	A concise account of findings, bias, applicability, and the assumptions relevant to triangulation.
Bias type — offset versus scaling	Offset (additive / shift) bias adds a distortion and can move a true null. Scaling (proportional / stretch) bias multiplies an effect on the stated scale; a true zero remains zero on that scale.

Alignment is not indirectness The two judgements answer different questions at different levels, and the stream-level estimand is what separates them. Alignment is a question of estimand identity, judged between two specified quantities: did this stream estimate the review's target quantity, or a related but distinct one? It is assessed in the narrative record below the SET — how it is recorded. Indirectness is a within-stream judgement: given the quantity the stream targets, does the evidence it contains apply to the population, setting, and exposure realization that its own estimand specifies? It is recorded in the SET's indirectness column. The two judgements can come apart in both directions. A stream can estimate its own estimand cleanly yet estimate the wrong quantity for the review (the TAAR1 animal behavioural outcome); another can target the right quantity yet draw on evidence that falls short of it — trials restricted to young, non-chronic adults for a target population that includes older people (A5).

Reference appendix

Additional methodological terms

These definitions retain the rest of the source glossary for reference.

Term	Working definition
Node	A variable in the diagram, such as exposure, a mediator, or an outcome.
Edge / arrow	An assumed direct causal influence between two nodes. A target question can concern several paths rather than one arrow.
Direct versus total effect	A direct effect excludes pathways through specified mediators; a total effect includes pathways through them. A trial's total-effect estimate should not automatically be attached to a direct arrow.
Confounder	A variable that causally influences both exposure and outcome, creating a non-causal route between them.
Effect modifier	A characteristic across whose values an effect differs on a specified scale. It is not necessarily a confounder.
Mediator	A variable on a causal pathway, such as exposure → mediator → outcome. Adjusting for it can change the quantity being estimated.
Collider	A variable caused by two other variables. Conditioning on it can open a non-causal path.

Term	Working definition
Scoping mode versus hypothesis mode	Scoping mode uses the diagram to clarify the question and evidence contributions. Hypothesis mode makes proposed pathways explicit for examination. Record the diagram's purpose and when it was introduced.
SMD — standardised mean difference	An effect expressed in standard-deviation units. Similar units do not make different outcomes or target quantities equivalent.
Bias direction — towards or away from the null	How a bias is expected to move an estimate relative to the true value, for a specified contrast and scale. Direction may be uncertain or involve crossing the null.
Triangulation	Drawing an interpretation across evidence contributions while examining their different and shared key sources of bias and assumptions.
Orthogonality / differing bias structures	The extent to which key bias mechanisms differ across streams. Independence needs justification; different design labels alone do not establish it.
Causal-assumption framework ("scaffolding")	The proposed structure connecting the DAG, explicit assumptions, evidence summaries, meeting deliberation, and reporting.
Assumption types	Four proposed categories: structural, identifying, proxy, and indirectness/applicability, defined below.
Structural assumption	A claim about a causal relationship or its absence, such as the proposed dopamine pathway in TAAR1 (A3).
Identifying assumption	A condition needed for a study's estimate to identify the stated causal quantity, such as sufficient confounding control for an observational contrast.
Proxy assumption	A claim that a measured proxy is informative about the target construct, such as the contested TAAR1 animal-to-human connection (A1).
Indirectness assumption	A claim about the extent to which findings transfer to the target population, intervention/exposure, or setting.
Target relationship / causal query	The question the review seeks to answer. It may concern an association, a total effect, a direct effect, or a mediated pathway.
Evidence stream	Evidence grouped by a relevant design or inferential approach; distinct identifying assumptions can require separate streams.
Evidence contribution map	A table showing what each stream can inform and which assumptions connect it to the target question.
Decision log	A dated record of changes, reasons, and ownership, linking successive diagram and register versions.
Edge status	The post-synthesis status of a relationship in the causal graph: **supported**, **unsupported**, **uncertain**, or **not adequately investigated**. Recorded in the decision log when the diagram is revised, paralleling the status judgements in the assumption register.
Evidential contribution type	The role a stream plays relative to the target: direct target contribution, related effect, mechanistic, contextual, or assumption-testing. Classified separately from alignment

Term	Working definition
	and risk of bias; a stream may hold several roles, and roles are not tied to particular study designs.
Target version	A dated, versioned statement of the review's target quantity or question. Amended only when the question is intentionally redefined, with justification and a record of the evidence known at the time. An evidential challenge to a mechanism does not by itself change the target.
SEM — Summary of Evidence on Mediators	The synthesis output of the hypothesis-generating route: a structured summary of evidence on candidate mediator relationships, from which a consensus meeting selects the mediators carried forward. A table, not a diagram — and not a structural equation model.
Causal graph	Umbrella term for DAGs, causal DAGs, path diagrams and related visualisations of assumed relationships between variables. An arrow is a causal claim only when it is given a causal interpretation; a diagram drawn to display associations or statistical model structure functions as a causal graph only under that reading.

Blank template

Download the assumption register (website download)

Proposal 1 · Update protocol templates

Why update the templates?

There are currently 2 GALENOS LSR Protocol Templates; 1) ; 2) The associations workflow produces a **Summary of Evidence table (SET)**. The hypothesis-generating workflow already recommends a DAG and produces a **Summary of Evidence on Mediators (SEM)**. The amendments connect the causal diagram, assumption register, and meeting record across both routes.

The retrospective TAAR1 example motivates the proposed changes:

Gap illustrated by the retrospective example	What the amendment addresses
Association and mechanism questions can occur in the same review.	Declare the review mode and the diagram's purpose.
Pathways may be reconstructed after the findings.	Record the proposed structure and challenge conditions before synthesis.
The two template routes produce different evidence summaries.	Link both routes to the DAG and assumption register.
Meeting reasoning may remain implicit.	Preserve the assumptions, judgements, and revisions in a reusable record.

What changes in each template?

The uploaded drafts frame all additions as recommendations. These practical prompts are a main methods contribution for review.

Template	Where the additions appear	Practical change proposed
Associations v3.2 draft	Research questions; reporting bias; evidence summary; triangulation meeting.	Declare the review mode, introduce a versioned DAG and register, record bias direction, and use the assumptions in meeting conclusions.
Hypothesis-generating v2.2 draft	DAG guidance; data extraction; SEM; bridge to triangulation; consensus meeting.	Version the existing DAG, extract assumption-relevant information, connect the SEM to the register, and prepare a SET-style input for triangulation.

Verbatim yellow-highlighted extracts — for template owners

The 25 highlighted paragraphs are reproduced without rewriting. The original Word drafts remain internal; a download of the proposed wording is provided below.

Associations v3.2 draft

Research questions

Review mode and causal structure (recommended):

We recommend that you state the primary mode of the review: evaluating associations, hypothesis-generating, or a mixture of both. If any research question concerns mechanisms of action or hypothesised causal pathways, we recommend declaring this explicitly as a secondary (hypothesis-generating) mode.

We recommend that, in that case, you draw a scoping directed acyclic graph (DAG) at protocol stage: the variables of interest, the hypothesised directional links between them (including candidate mediators), and the outcome(s). The DAG is a working hypothesis, not a finding, and should be versioned (dated and archived).

We recommend that research questions about causal pathways are then answered edge by edge against the pre-specified DAG (i.e. "is the evidence consistent with this hypothesised link?"), rather than by reconstructing pathways after the findings.

Reporting bias

We recommend that, for each major bias, the likely direction of the resulting error is recorded relative to the relevant DAG edge (towards null, away from null, or unpredictable), in addition to the risk-of-bias judgement. Where quantitative bias adjustment is planned, the bias type can also be recorded: offset (additive) bias shifts every estimate by a roughly fixed amount and can create an effect where the true effect is zero, whereas scaling (proportional) bias inflates or shrinks estimates in proportion to their size and cannot.

Summary of the evidence

Making the Summary of Evidence (SET) table DAG-compatible (recommended):

We recommend adding the following fields to each row of the Summary of Evidence (SET) table: (i) which DAG edge(s) the evidence stream informs; (ii) up to three key assumptions on which the estimate depends, each typed as structural, identifying, indirectness or proxy, with status (agreed / contested / uncertain); (iii) the expected direction of each major bias relative to the relevant DAG edge (towards null / away from null / unpredictable); and (iv) a link to the versioned DAG and assumption register.

We recommend that the assumption register is seeded at protocol stage and versioned. Each entry records the assumption, its type, the evidence streams it concerns, its status, and what evidence would challenge it.

Triangulation meeting

Using the DAG and assumption register at the triangulation meeting (recommended):

We recommend that, before the meeting, each SET table row is mapped onto the DAG edge(s) it informs. Streams that inform different edges do not directly agree or disagree; any statement about agreement between them traverses an assumption (e.g. a proxy assumption) that should be named.

We recommend that conclusions are reported per DAG edge, with the traversed assumptions stated, rather than as a single global judgement.

We recommend that orthogonality is judged from the directions of the streams' major biases: streams whose biases favour findings in opposite directions (one towards null, one away from null) make agreement between them strongly informative. Representing each bias as a factor in the DAG makes this a checkable claim (the biases should share no common cause).

We recommend applying three revision triggers during the meeting: (a) evidence incompatible with a DAG edge; (b) an evidence stream that maps onto no edge; (c) an assumption challenged in discussion. Each trigger should lead to a versioned update of the DAG or assumption register.

Hypothesis-generating v2.2 draft

DAG and assumption register

Versioning the DAG and attaching an assumption register (recommended):

We recommend that the DAG is treated as a versioned working model: it will be refined as evidence accumulates, and each version is dated and archived.

We recommend that, for each edge, you note which evidence streams (non-human, human observational, human interventional, mechanistic) can inform it.

We recommend attaching an assumption register to the DAG. For each assumption, record: its type (structural — an asserted causal link; identifying — what makes a stream's estimate causal; indirectness — how closely the stream's population, intervention and outcome match the target question, including animal-to-human relevance; proxy — a measured outcome standing in for the target outcome); its status (agreed / contested / uncertain); the streams it concerns; and what evidence would challenge it.

Data extraction

We recommend also extracting information relevant to the assumption register — for example the timing of mediator measurement, population characteristics, and intervention dose or exposure descriptors — so that indirectness and proxy assumptions can be re-evaluated as evidence accumulates, and not only effect estimates.

Summary of Evidence on Mediators (SEM)

Making the SEM DAG-compatible (recommended):

We recommend adding the following fields to the SEM for each candidate mediator: (i) which DAG edge(s) the evidence about it informs; (ii) up to three key assumptions (typed, and with status, as in the assumption register); and (iii) a link to the versioned DAG and assumption register.

Bridge to triangulation

Bridging to triangulation (recommended):

We recommend that, because this template produces a Summary of the Evidence on Mediators (SEM) rather than a Summary of Evidence (SET) table, teams preparing for triangulation derive a one-row-per-stream SET-style table from the DAG (using the fields described in the associations template) so that both template routes feed the triangulation meeting comparably.

Consensus meeting

Reviewing the assumption register at the consensus meeting (recommended):

We recommend that the assumption register is a standing agenda item: assumptions challenged by the accumulated evidence are re-statused (agreed / contested / uncertain), and the DAG is revised if a revision trigger fires (evidence incompatible with an edge; a stream that maps onto no edge; an assumption challenged in discussion).

We recommend that conclusions are reported per DAG edge, naming the assumptions they traverse. Note that a small mechanistic study that measures a node on a DAG edge can have probative value out of proportion to its precision, because it bears on whether an association is causal rather than confounded.

Specifying the target quantity

The amendments above make the causal structure explicit; a companion step makes the *quantity* explicit. Companion guidance (Component A) helps a team specify what each question targets before eligibility and extraction are fixed. Questions are classed as **descriptive, associational, causal or hypothesis-generating** — a review can mix modes, and the mode does not determine the template route. The operational rule: a **causal-mode question in the associations route triggers the DAG and register amendments**, because that template contains no DAG by default; mechanistic or hypothesis-generating questions use the hypothesis route and link the SEM to the register. The specification extends PICO/PMICO rather than replacing it — each worksheet step maps to a familiar element, with the summary quantity, intercurrent events, effect type and evidence streams as explicit extensions.

The guidance also specifies a **stream-level estimand** for each evidence stream — the quantity that stream can estimate in its own terms — so that alignment at the evidence-summary stage relates two specified quantities rather than a study and an underspecified ambition.

These stream-level estimands are also the reference point for within-stream indirectness: the SET's indirectness column judges how far each stream's evidence applies to the population, exposure, outcome, and timing that its own estimand specifies. This is what keeps the two judgements non-overlapping — indirectness stays within the stream; alignment relates the stream's quantity to the review target.

The TAAR1 page shows an illustrative completed target statement.

Specification worksheet, stream-level estimands, and governance

A3. Specification worksheet — review-team prompts

Copy this section into a review-specific note. Complete provisionally, discuss ambiguities, and revise after causal-structure discussion.

Step	Signalling question	Provisional answer / unresolved issue
1. Purpose	What decision or scientific explanation motivates the review?	*To complete*
2. Mode	Descriptive, association, causal, hypothesis-generating or mixed? Why?	*To complete*
3. Target population	Who or what should the answer apply to? Setting and eligibility?	*To complete*

Step	Signalling question	Provisional answer / unresolved issue
4. Exposure/intervention	What are the relevant level(s), versions, duration and implementation?	*To complete*
5. Comparator	Compared with what observable or hypothetical alternative?	*To complete*
6. Outcome/construct	What is the *target construct*, and by what measure and direction?	*To complete*
7. Timing	When are exposure, mediator (if any) and outcome assessed?	*To complete*
8. Summary quantity	Risk difference/ratio, mean difference, prevalence, association slope, etc.?	*To complete*
9. Intercurrent events	Which events change interpretation/ existence of outcome? Which strategy addresses them (if applicable)?	*To complete*
10. Effect type	Total effect? A specified direct or indirect effect? Conditional or marginal?	*To complete*
11. Evidence streams	Which streams might estimate the target? Which could inform related effects, pathways, context or assumptions?	*To complete*
12. Assumptions	What must be true to interpret or transport stream findings to this target?	*To complete*
13. Open questions	What still needs expert, stakeholder or methodological input?	*To complete*

Writing the target statement

- **Causal template:** "In [target population], what is the [summary] of [outcome at time] under [intervention strategy] compared with [comparator strategy], treating [key intercurrent events] using [specified strategy]?"
- **Association template:** "In [population and period], what is the [marginal/conditional association measure] between [defined exposure] and [outcome at time], with [conditioning variables/standardisation]?"
- **Descriptive template:** "In [population/period], what proportion/mean [well-defined property] is observed, measured by [definition]?"
- **Hypothesis-generating template:** "Which candidate variables or pathways might connect [exposure] and [outcome], and what forms of evidence could support or challenge the proposed relationships?"

These are **templates for clarification**, not guarantees of identification.

Specifying stream-level estimands (A3a)

For each evidence stream nominated in worksheet step 11, specify the quantity that stream can estimate *in its own terms*. Illustrative TAAR1 application (placeholders to be confirmed against the included studies; not a completed assessment):

Field	Human stream (TAAR1, illustrative)	Animal stream (TAAR1, illustrative)
Population	Adults with schizophrenia spectrum disorders in the included trials	The strains, sexes and challenge paradigms used in the included experiments
Intervention/exposure	Add-on TAAR1 agonist at trial doses	Compound at preclinical doses in a challenge-induced state
Comparator	Placebo add-on (or as reported)	Vehicle under challenge

Field	Human stream (TAAR1, illustrative)	Animal stream (TAAR1, illustrative)
Outcome/construct	PANSS total (a measure of the symptom construct)	Reversal of challenge-induced hyperlocomotion (a behavioural outcome)
Timing	4–6 weeks	As reported per paradigm
Summary quantity	As reported (e.g. mean difference, SMD)	As reported (e.g. SMD)
Effect type	Assignment effect under the trials' strategies (to confirm)	Experimental contrast under each study's design

The proxy assumption A1 is then a **bridge between two specified estimands** — a claim that the animal stream's quantity is informative about the human stream's construct — rather than a vague link between "animal studies" and "the human question".

Governance of the target statement (proposal)

- **Drafting:** the review team drafts the target statement and stream-level estimands at protocol stage, using the worksheet.
- **Sign-off:** agreed with clinical, preclinical and methodological input before eligibility screening begins; experiential and stakeholder input informs outcomes and contrasts.
- **Amendment:** any change after sign-off requires a version increment, a dated justification, and a note of the evidence known at the time.
- **Disagreement:** record disagreement and the positions held in the register rather than forcing consensus; unresolved disagreement is carried into triangulation as an explicit condition on interpretation.

These arrangements are proposals for discussion; GALENOS governance groups should confirm ownership and process.

Download the full estimand specification guidance (.md) (website download)

Link to the evidence summary

The **six existing SET columns stay unchanged**. One added **Key Assumptions** column records the relevant relationships, typed assumptions and status, and links to the versioned DAG and register. Bias direction remains in the existing bias columns. See the worked SET and template.

For a review already underway, record when the framework was introduced and which evidence was already known. LSR8 illustrates this timing distinction.

The four questions on the Briefing page guide the meeting discussion.

Proposed wording download

Highlighted wording only (.md) (website download)

Proposal 2 · Triangulation meeting worksheet

Purpose

The worksheet gives the triangulation panel a common record of how evidence changes its interpretation of the DAG and its assumptions. The **Summary of Evidence table (SET)** summarises the evidence; the **assumption register** records the conditions attached to the interpretation.

See the TAAR1 DAG and register for the worked illustration. The worksheet below turns that reasoning into prompts a meeting can use.

Four steps

Step	What the team records
1. Map	The relationship or query each stream informs, the estimate or finding, and the assumptions needed to connect it to the target question.
2. Compare	Whether each stream addresses the target causal question, a related question, or a different link (causal alignment assessment); which integration level that supports — A common effect, B related but non-identical, C different links; where findings agree, diverge, or complement.
3. Examine bias	Shared and differing bias mechanisms, their expected directions, magnitudes, and uncertainty, and the implications for interpretation.
4. Revise and record	The conclusion and its conditions; edge statuses for revised relationships (supported / unsupported / uncertain / not adequately investigated); changes to assumptions or the diagram; unresolved questions and actions.

Proposed worksheet fields

Before the discussion	After the discussion
Target question and **target specification version**; date and participants; diagram, register, and SET versions; evidence already known when the framework was adopted.	Conclusion and conditions; integration level chosen and its rationale; edge statuses for revised relationships; revised versions; actions, owners, and review dates.

Causal alignment assessment within the four steps

Companion guidance (Component B) supplies the questions and records *within* each step — no new procedure is needed. Each evidence stream gets a **stream-level estimand** (the quantity it can estimate in its own terms) prepared before the meeting, so Step 2 compares specified quantities rather than a study with an underspecified ambition.

Step	Component B support	Artefact produced
Preparation	Retrieve the versioned target specification; draft stream-level estimands.	Draft stream-level estimands in the meeting pack
1. Map	Place each contribution on the edge, node, proxy connection or assumption it informs — not automatically a single arrow; classify contribution type.	Contribution map with register IDs
2. Compare	Alignment discussion: did streams estimate the same *quantity*? Resolve the integration level (A common effect / B related but non-identical / C different links). Distinct from the	Below-SET narrative record per stream (draft)

Step	Component B support	Artefact produced
	SET's indirectness column (within-stream applicability).	
3. Examine bias	Identification and within-stream indirectness; then the orthogonality check — shared vs differing bias mechanisms relative to each stream's own estimand.	Orthogonality judgement; bias notes keyed to register IDs
4. Revise and record	Conditional interpretation; route each change as alignment re-interpretation, DAG/register update, or formal target amendment.	Final narrative records; versioned updates

Steps 1 and 2 **iterate** rather than sequence strictly: alignment assessment comes before or alongside mapping, never after cross-stream agreement is declared.

Alignment questions, descriptors, narrative record, and orthogonality check

Opening alignment questions (B1a)

- What exactly was the review target, and which version of that specification is being used?
- What did the included studies aim to estimate, and can that be inferred confidently from the reports?
- What does the synthesis summarise: a common effect, a distribution of study effects, an association, a qualitative pattern, or several distinct results?
- What quantity did *this stream* target in its own terms (the stream-level estimand)? Alignment is judged between two specified quantities, not between a study and an underspecified ambition.
- Does the evidence provide an estimate of the target or a contribution that must be interpreted through another assumption or causal pathway?
- Could an apparent agreement or discrepancy arise because **different quantities** were estimated rather than because studies yielded conflicting evidence?

When the review has no numerical target estimand (hypothesis-generating mode): the alignment question shifts from *does this quantity match the target quantity?* to *does this evidence inform the proposed relationship, its measurement, or the assumptions connecting its components?* The SEM's rows are assessed against pathway components rather than a pooled effect.

Dimension-specific signalling questions (B1b)

These dimensions compare the stream-level estimand with the review target — a between-level judgement, recorded in the alignment narrative. They do not absorb within-stream questions: a shortfall of a stream's evidence against its own estimand — for example, trials that under-represent the population its estimand specifies — is indirectness, and belongs in the SET's indirectness column, not here.

Dimension	Signalling questions (examples)
Population / setting	Who is represented by the study and synthesis? Who is in the target? Are age, diagnosis, species, severity, context or effect modifiers different?
Intervention / exposure / comparator	Are dose, timing, treatment versions, assignment versus receipt and comparison conditions the same? Is a hypothetical intervention well-defined?
Outcome / construct / measurement	Is the outcome the target construct, a different manifestation or a proxy? Could changes in measurement alter direction or meaning?
Timing	Do exposure period, outcome follow-up, mediator order and treatment duration match?
Quantity / causal effect definition	Is the target descriptive, associative or causal? Total effect, assignment effect, hypothetical adherence effect, direct or mediated effect?

Dimension	Signalling questions (examples)
Synthesis target / summary measure	What does the pooled estimate or SWiM summary target? Are measures transformable? Does pooling over studies yield the target-population quantity?
Intercurrent events (where relevant)	How were discontinuation, switching, rescue treatment or competing events handled? Is this compatible with the target?

Descriptive labels (B1c) — provisional language only

- **Closely aligned:** the stream addresses a compatible target quantity, notwithstanding methodological validity and applicability questions.
- **Partly aligned:** substantial correspondence, but a consequential difference or qualification remains.
- **Different but relevant:** the stream addresses a distinct quantity, yet can inform the target through a specified mechanism, proxy, context or assumption.
- **Connection unclear:** reporting or conceptual uncertainty prevents a defensible judgement.
- **No established link to this target:** currently no justified inferential connection; this does not mean the evidence has no other scientific value.

These are descriptors, not categories with validated thresholds, and should not be forced into a global rating.

From alignment judgements to the integration level

The alignment judgements resolve into the **integration level** for each pair or set of streams, which determines how they may be brought together:

- **A — Sufficiently common causal effect:** streams estimate effects defensibly related to one target estimand → quantitative integration may be possible (transformations, transportability assumptions, bias adjustment).
- **B — Related but non-identical effects or constructs:** streams address closely related questions but cannot defensibly be represented as observations of one target effect → retain within-stream quantitative synthesis; compare across streams without pooling.
- **C — Different links in a broader causal explanation:** streams inform different edges or propositions in the causal model → complementary, pathway-/question-level triangulation; "agreement" is not the operative concept.

The level is a recorded judgement with its rationale, not a score; borderline or mixed cases (for example, closely aligned on intervention, different on outcome) should be described per dimension as above.

Blank narrative record (below the SET table, one per stream row)

SET row and target reference: [review, outcome, target version, synthesis version].

Stream-level estimand: [what this stream targeted in its own terms].

What is being summarised? [one to three sentences on study-level variation and the synthesis target].

Main points of alignment or difference: [key population, contrast, outcome, time and effect-definition considerations].

Why this evidence contributes: [direct/related/mechanistic/contextual/assumption-relevant contribution].

What must be assumed to connect it to the target? [DAG/register references].

Unresolved questions for triangulation: [reporting gaps, validity, transportability, bias and possible challenges].

Orthogonality check (Step 3 support)

- List the key bias mechanisms for each stream (selection, measurement, reporting, confounding, analysis) relative to each stream's *own* estimand.
- Identify mechanisms that could operate in **both** streams — shared publication practices, shared measurement instruments, shared populations, or animal-to-human assumptions that transmit bias in the same direction.
- Ask: if the shared mechanism operated, would it produce the observed pattern of agreement or disagreement?
- Record whether the differing-bias premise of the triangulation is *justified for this pair of streams*, or merely asserted by design labels.

Agreement between streams with plausibly shared bias should be reported as conditional in the same way as agreement resting on a contested proxy assumption.

Download the full estimand alignment guidance (.md) (website download)

The Word worksheet provides blank tables and response spaces for these fields and the four steps. It can be completed before and during a meeting, then kept with the review records.

Detailed questions for follow-up

- Which fields are essential for the panel, and which can be prepared or maintained elsewhere?
- Is the preparation and recording burden justified, and who owns the record between updates?

The four questions on the Briefing page guide the meeting discussion.

Blank templates

Download the fillable Word worksheet (.docx) (website download) Markdown source (.md) (website download) Assumption register (.csv) (website download)

Proposal 3 · Reporting and the Summary of Evidence table

The idea

The **Summary of Evidence table (SET)** keeps its existing columns for results, bias, indirectness, and other considerations. The proposed addition is a **Key Assumptions** column containing short typed statements or IDs, their status, and links to the DAG and assumption register. The register holds the detailed reasoning.

Worked example: TAAR1

The retrospective example shows why the measured animal outcome and the human symptom outcome need separate rows. Their relationship is the contested **proxy assumption A1**, shown as a dashed connector in the TAAR1 diagram.

The six existing SET headings are retained. **Key Assumptions** is the only added column, shaded to distinguish it from the original fields.

Source of evidence	Summary of association	Within-study bias	Bias due to Missing Evidence	Indirectness	Other considerations	Key Assumptions
Human randomised trials	PANSS total, 4–6 weeks; agonists versus placebo. **SMD 0.15 (−0.05 to 0.34)**; 4 trials, 1,291 participants.	Assessed low risk in the retrospective example.	Assessed low risk in the retrospective example.	Assessed low overall; population and measurement relevance remain contested.	COVID-era trial context may affect placebo responses.	Treatment → human symptoms. A4: identifying, agreed; A5: indirectness, contested; A6: proxy, contested. DAG · Register.
Animal experiments	Locomotor activity; agonists versus control. **SMD 1.01 (0.74 to 1.27)**; 125 comparisons.	Moderate risk; expected to exaggerate the effect.	Moderate risk; possible publication bias may exaggerate the effect.	Moderate; proxy and exposure applicability uncertain.	Prevention versus reversal of the induced phenotype remains unresolved.	Treatment → animal outcome. A1: proxy, contested; A2: proxy (bridge), contested; A7: structural, uncertain. DAG · Register.

Proposed meeting wording: “The streams inform different outcomes. Their combined interpretation for human symptoms depends on the relevance of the animal proxy and the bias and applicability of each contribution.”

Template interpretation and assumption IDs follow the retrospective worked example as described in Smith et al. 2025 and Siafis et al., 2024. This table presents the evidence used for the retrospective example, not a current review update.

Proposed template

Keep the six existing column headings from the uploaded associations template. Add **Key Assumptions** as the final column, shaded below. Use one row per evidence stream for the outcome under discussion.

Source of evidence	Summary of association	Within-study bias	Bias due to Missing Evidence	Indirectness	Other considerations	Key Assumptions
Review / evidence stream.	Outcome, contrast, estimate and uncertainty, or qualitative summary.	Within-study risk of bias, critical appraisal and expected direction.	Reporting-bias assessment and expected direction.	Applicability of the stream's evidence to the population, species, model, exposure, or outcome specified in the stream's own estimand.	Relevant issues not covered elsewhere.	Relationship informed; up to three typed assumptions and status; links to the versioned DAG and register.

Below the table: state the meeting conclusion, the assumptions it depends on, and unresolved questions. The existing bias and indirectness columns retain their roles; the added column connects those judgements to the causal model.

Alignment is recorded below the table, not in it. The Key Assumptions column holds only the relationship informed, categorised assumption IDs with status, and links. Below the table, one short narrative record per stream row asks how closely the evidence stream estimated the review's target estimand — a question of estimand identity, judged between two specified quantities: the stream-level estimand and the review's target estimand. This is a different question, from indirectness, which is a within-stream judgement: given the quantity the stream targets, does the evidence it contains apply to the population, setting, and exposure realization that its own estimand specifies? The two judgements can come apart in both directions. An evidence stream can estimate its own estimand cleanly yet estimate the wrong quantity for the review (the TAAR1 animal behavioural outcome), or target the right quantity yet draw on evidence that falls short of it — trials restricted to young, non-chronic adults for a target population that includes older people.

Worked alignment narrative — TAAR1 animal stream

Illustrative narrative record — TAAR1 animal stream (retrospective reconstruction, for calibration only; not a completed assessment):

> **SET row and target reference:** TAAR1 animal experimental evidence; target version [draft specification, Component A §A5].
>
> **Stream-level estimand:** Effect of [compound] versus vehicle on reversal of challenge-induced hyperlocomotion in [strains/paradigms as included], reported as [SMD].
>
> **What is being summarised?** Experimental contrasts across animal studies; the synthesis summarises a distribution of animal-specific effects, not the human clinical quantity.
>
> **Main points of alignment or difference:** Different population (species, challenge state), different intervention (preclinical dosing), different outcome (locomotor behaviour versus PANSS-assessed symptoms), different effect definition (experimental contrast, not a treatment-policy effect).
>
> **Why this evidence contributes:** Related-effect and mechanistic contribution; informs the proposed pathway and the behavioural-proxy assumption rather than the clinical target directly.
>
> **What must be assumed to connect it to the target?** A1 (behavioural proxy validity, contested) and A2 (dose/compound applicability, contested).
>
> **Unresolved questions for triangulation:** Whether proxy validity can be evaluated empirically; whether shared biases (for example, publication practices or selective outcome reporting operating across species) could produce concordant directions.

Download the full causal alignment assessment guidance (.md) (website download)

Reporting the meeting outcome

The proposed wording makes the conditional interpretation explicit. For TAAR1, the key point is that a judgement about human symptoms cannot treat the animal proxy as the same measured outcome.

Situation	Suggested sentence structure
Contributions support the same interpretation	"Evidence from [streams] is consistent with [interpretation], conditional on [assumptions]."
Agreement may reflect shared bias	"The findings agree, but shared bias through [mechanism] remains a possible explanation."
Related but non-identical effects (integration level B)	"The [animal] and [human] streams address related but non-identical outcomes; their estimates are compared but not pooled, and their consistency is conditional on [assumption]."
Complementary contributions (integration level C)	"The streams inform different links in the causal pathway and are interpreted as complementary; together they [support / leave uncertain] the proposed explanation, conditional on [assumptions]."

Situation	Suggested sentence structure
Contributions diverge	“The findings diverge for [relationship]; differences in [estimands, applicability, or bias] remain unresolved.”

Ensuring magnitude and uncertainty. Conclusions about convergence or divergence must preserve or point to effect direction, magnitude and uncertainty, not direction alone; a summary estimate should not be reduced to a qualitative claim such as “supports efficacy”.

Detailed questions for follow-up

- Does the “Key Assumptions” column provide enough information without crowding the existing table?
- Which wording examples need design-specific guidance e.g. animal studies, RCT, interrupted time series?

Blank template and wording

Download the blank SET (website download) Download draft wording (website download)

TAAR1 · Retrospective application

The diagram, register, worksheet framing, and reporting language were reconstructed after the review and meeting. They illustrate what the framework could make explicit; the original panel did not use these tools.

TAAR1 illustrates how human symptom outcomes, animal behavioural outcomes, and mechanistic evidence contribute to different parts of the question. This page holds the worked diagram and register; the Summary of Evidence table (SET) shows how they connect to the reported findings.

DAG and selected assumptions

A **proxy** is a measured variable used to stand in for another construct. Here **A1** marks the contested animal-to-human proxy connection and **A3** the uncertain proposed mechanism.

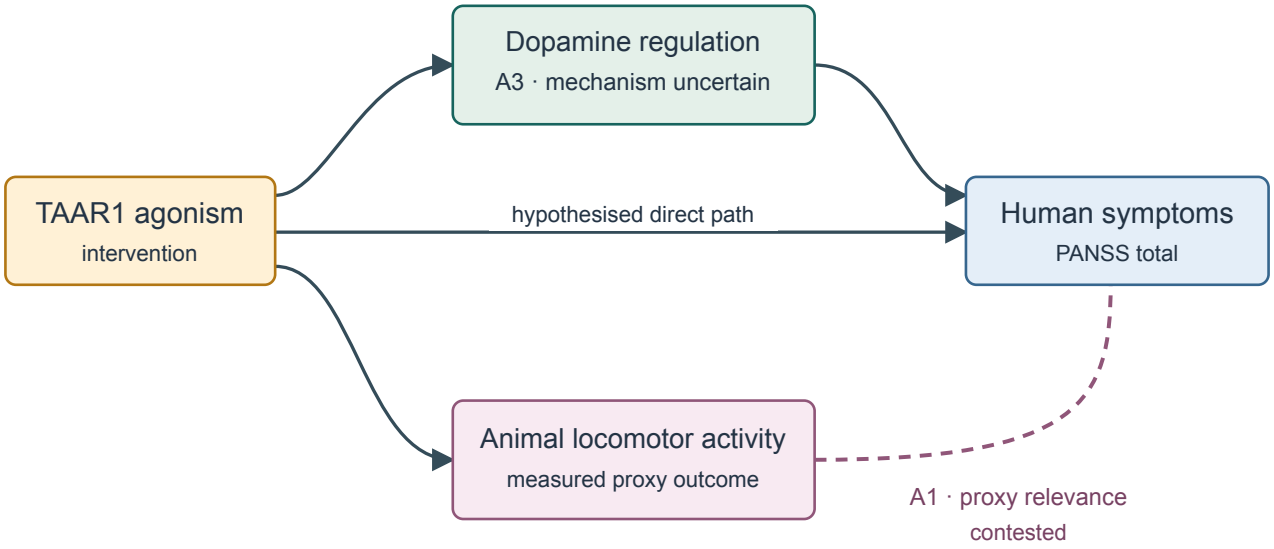


Figure 2: TAAR1 retrospective DAG with selected assumptions highlighted. Solid arrows show hypothesised causal paths; the dashed A1 connector represents proxy relevance rather than a causal arrow.

Assumption register

These summaries retain the source register's IDs and judgements. The labels connect the diagram to explicit statements that can be discussed and challenged.

ID	Type	Assumption	Status
A1	Proxy	Reversal of challenge-induced hyperlocomotion is informative about the target human symptom outcome.	Contested.
A2	Proxy (bridge)	Preclinical dosing and compounds correspond to human therapeutic exposure — the exposure-side bridge between the animal stream's estimand and the human target, alongside A1 on the outcome side.	Contested.
A3	Structural	TAAR1 agonism influences symptoms through presynaptic dopamine regulation.	Uncertain.
A4	Identifying	Randomisation and blinding support the human trial comparison.	Agreed.

ID	Type	Assumption	Status
A5	Indirectness	The trial populations adequately represent the population specified in the human stream's estimand.	Contested.
A6	Proxy / measurement	PANSS total is sensitive to the symptom changes relevant to the proposed mechanism.	Contested.
A7	Structural	The animal effect concerns reversal of an established phenotype rather than prevention of its induction.	Uncertain.

Evidence contributions — supporting detail

Contribution	What it informs
Human randomised trials	The treatment contrast for measured human symptom outcomes.
Animal experimental evidence	The effect on the measured animal behavioural outcome; human relevance requires a proxy assumption.
PET mechanistic substudy	The proposed dopamine-related mechanism; this does not by itself establish mediation or clinical benefit.

An inferential proxy connection is distinct from a physical causal arrow. Recording it does not make the animal and human outcomes the same estimand.

Target specification

The register above records assumptions; the target itself also needs stating. The retrospective materials do not contain a fully specified target, so the following is an **illustrative completed statement, for calibration only** — bracketed content is a placeholder to be confirmed against the actual protocol and trials, not a claim about them:

> In [adults with schizophrenia spectrum disorders], what is the [mean difference] in [PANSS total score] at [4–6 weeks] under [add-on TAAR1 agonism at the doses used in the included trials, continued as assigned] compared with [placebo add-on], treating [treatment discontinuation and switching] using a [treatment-policy strategy]?

Even this statement leaves open whether the review's interest is the assignment effect (as stated), a hypothetical fully-adherent effect, or a mechanistic pathway. The specification worksheet forces that decision into the open; the full guidance is available as a download (website download).

From the example to the templates

Use the protocol prompts to prepare the proposed structure, the four-step worksheet to record deliberation, and the worked SET and wording to report the conditional interpretation. The SET includes the actual estimates and 95% confidence intervals.

The retrospective example can expose ambiguous terms and missing fields. Practical usefulness and burden need testing in LSR8′s ongoing reviews.

This is an illustrative retrospective reconstruction. Evidence sources and limitations. The Briefing page holds the meeting questions.

Blank templates

Protocol amendments (website download) Word meeting worksheet (website download) SET template (website download)

LSR8 · Heat & Mental Health

The protocol is complete and the living systematic reviews are underway. The framework is being introduced **after protocol completion**; its adoption date and the evidence already known need to be recorded.

Working DAG: the human-only version

This is the original **human-only hypothesis diagram, figure 3, v1.1**. Orange nodes represent exposures, green nodes mediator groups, and blue nodes outcomes. The arrows show hypotheses to examine, not established effects.

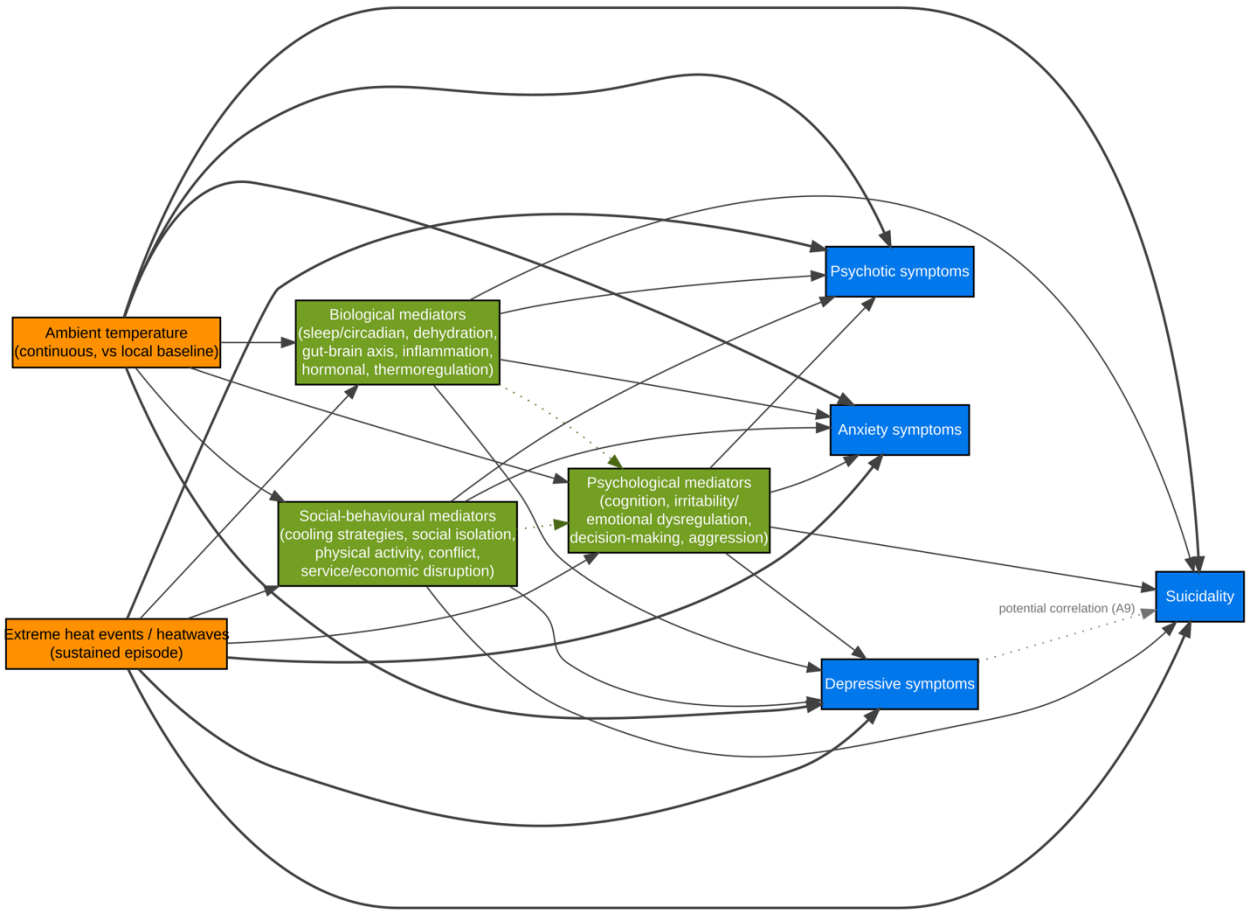


Figure 3: LSR8 human-only hypothesis DAG v1.1. Solid arrows show hypothesised causal influences; the dotted green connections are proposed mediator relationships, and the grey dotted line is a correlation annotation.

Priority adjudications

Issue	Why the decision matters
Medication use and social support	Their proposed roles may differ with temporal ordering; treating them as confounders, or shared causes of exposure and outcome, by default could change the quantity estimated.
Cooling-to-heat applicability (A7)	Determines the contribution cooling RCTs can make to the target heat question. Deciding what the target quantity is — an observed association or a defined intervention effect — is the kind of question the estimand specification guidance is designed to settle before synthesis.
Ambient temperature versus extreme heat (A3)	Determines whether the two exposure constructs require distinct comparisons and interpretation.
Animal proxy and duration eligibility (A1–A2)	Determines what animal evidence is eligible and how it could inform human outcomes.
Depression and suicidality (A9)	Requires clarity on whether a line represents a causal hypothesis or a descriptive association. A correlation annotation is not a causal DAG edge.

Applying the three methods guidance elements

As the **Summary of Evidence tables (SETs)** become available, use them to examine what each stream informs and which assumptions require revision.

1. **Establish the baseline:** record when the framework entered use, the available evidence, and which choices restate or extend the protocol.
2. **Use the worksheet:** record the interpretation, disputed assumptions, revisions, and actions through the four steps.
3. **Carry the judgement forward:** link the reported conclusion to the diagram, register, and evidence-summary versions used.

Supporting DAGs, selected assumptions, and evidence contributions

The five other original images support more detailed discussion. They are retained here as reference material.

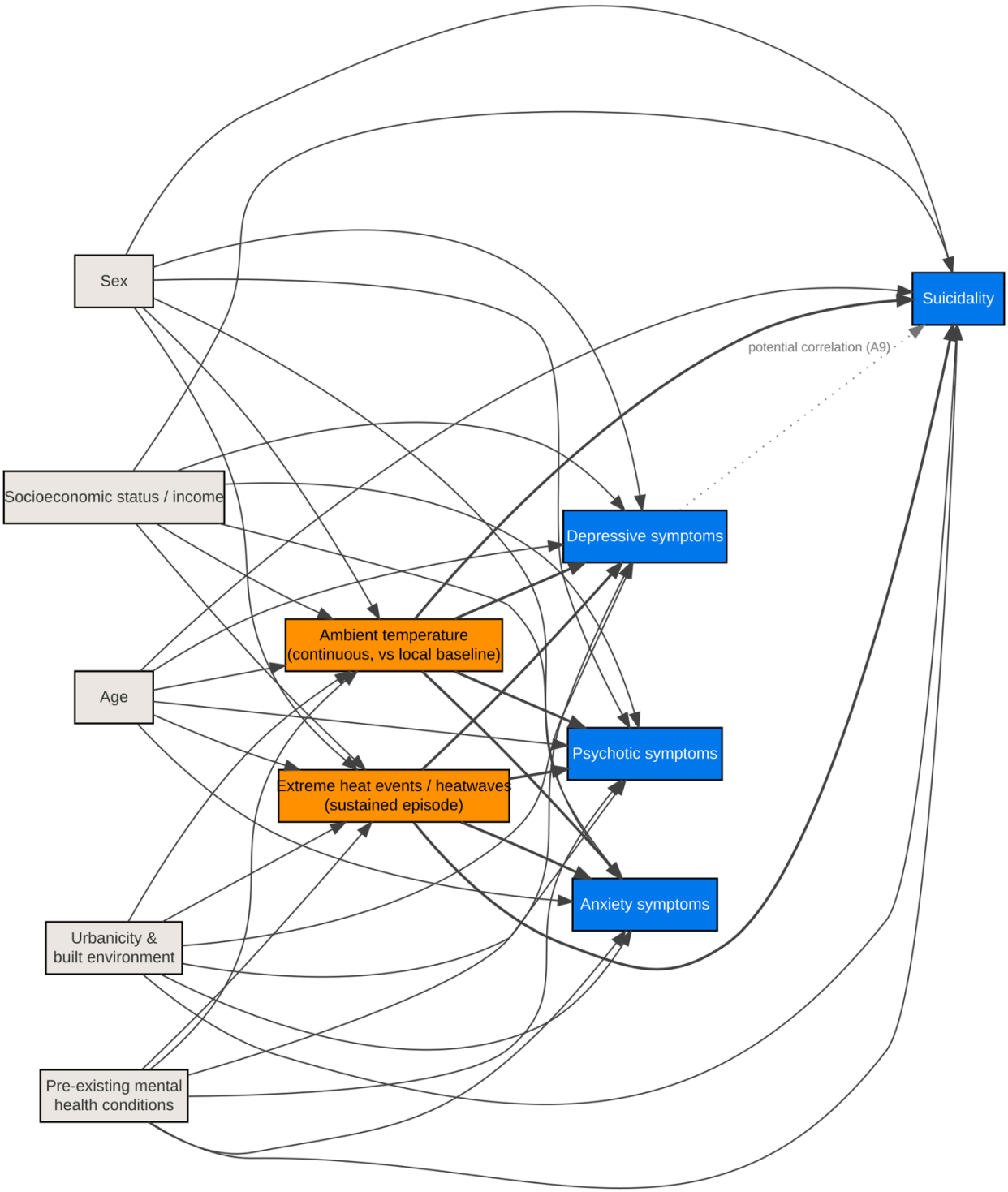


Figure 4: Scoping diagram v1.1: candidate confounders, two heat exposures, and four outcomes. Proposed covariate roles need adjudication.

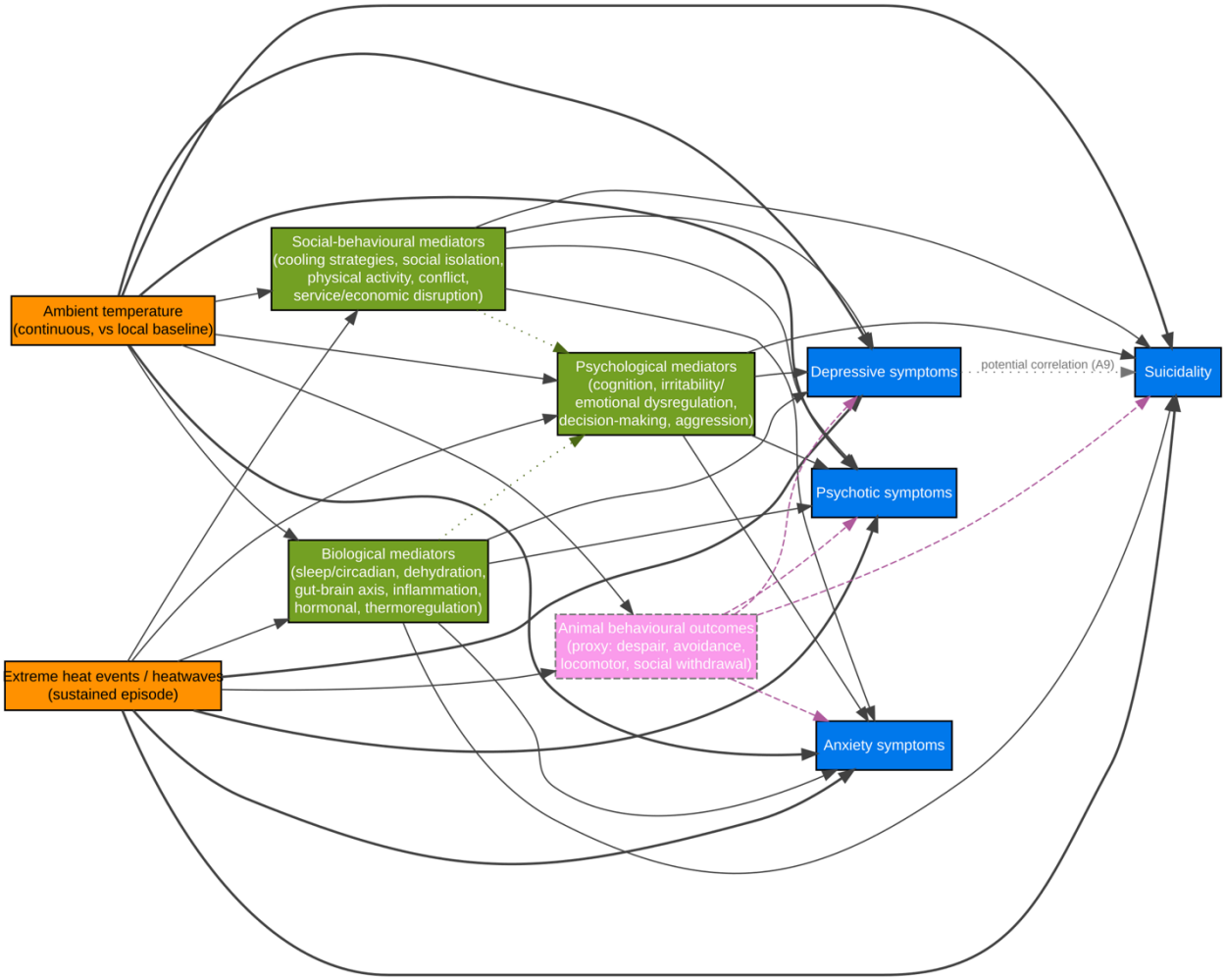


Figure 5: Hypothesis diagram with animal proxy v1.1. Pink dashed connections represent proxy assumptions A1 rather than physical causal effects of animal outcomes on human outcomes.

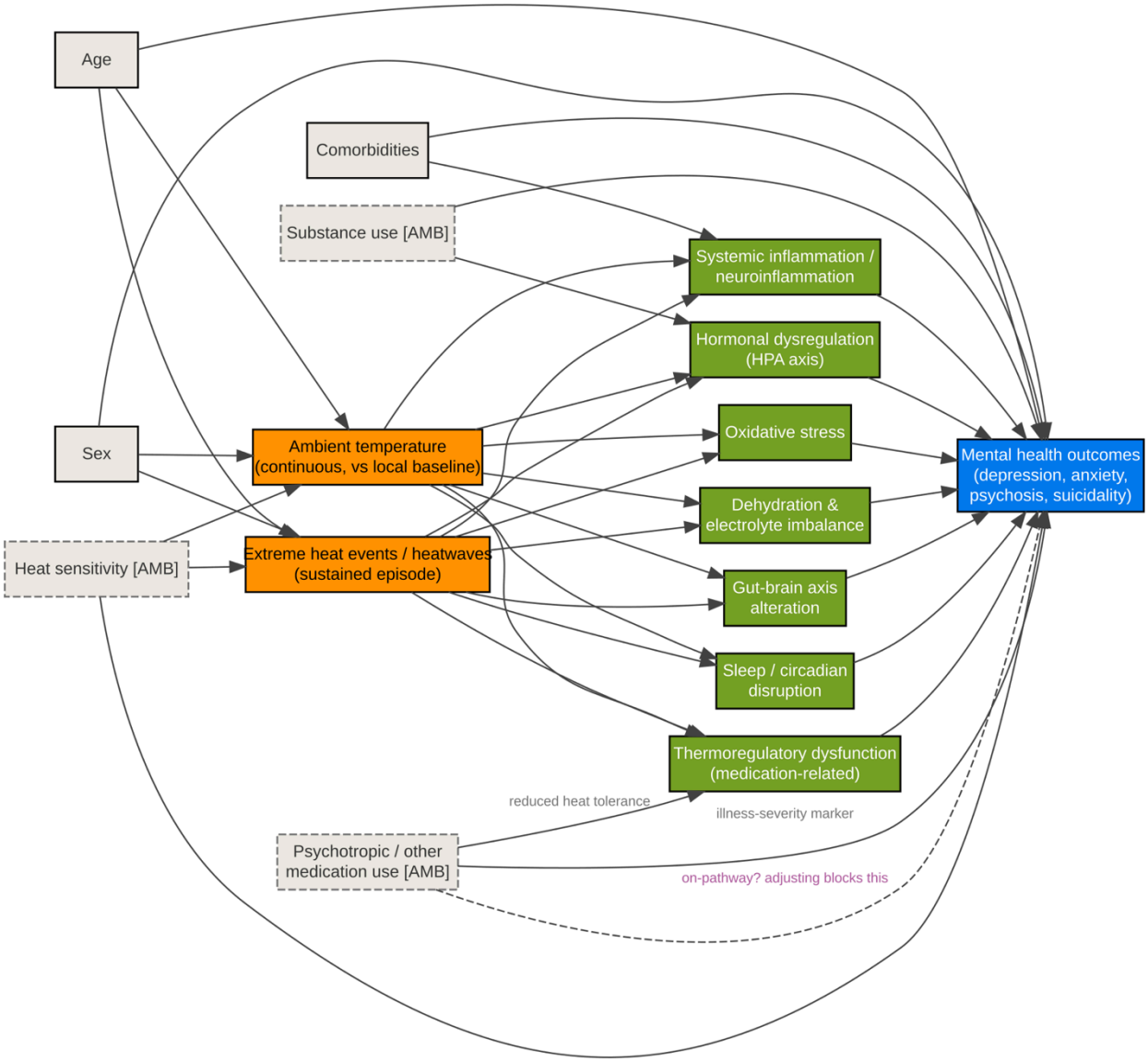


Figure 6: Biological mediator detail v1.1. Candidate variable roles, including medication use, require adjudication.

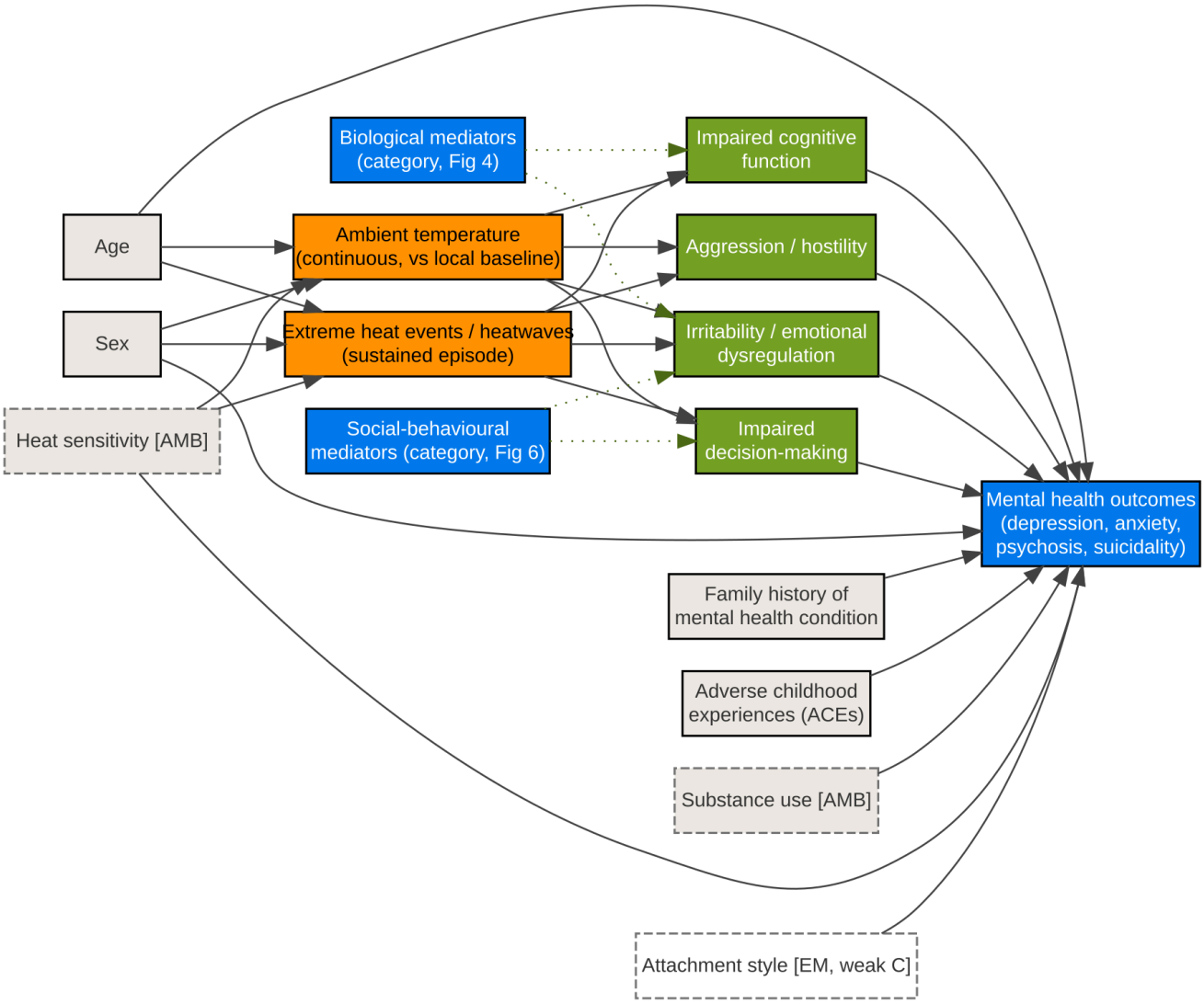


Figure 7: Psychological mediator detail v1.1. Proposed links from the other mediator groups remain open for discussion.

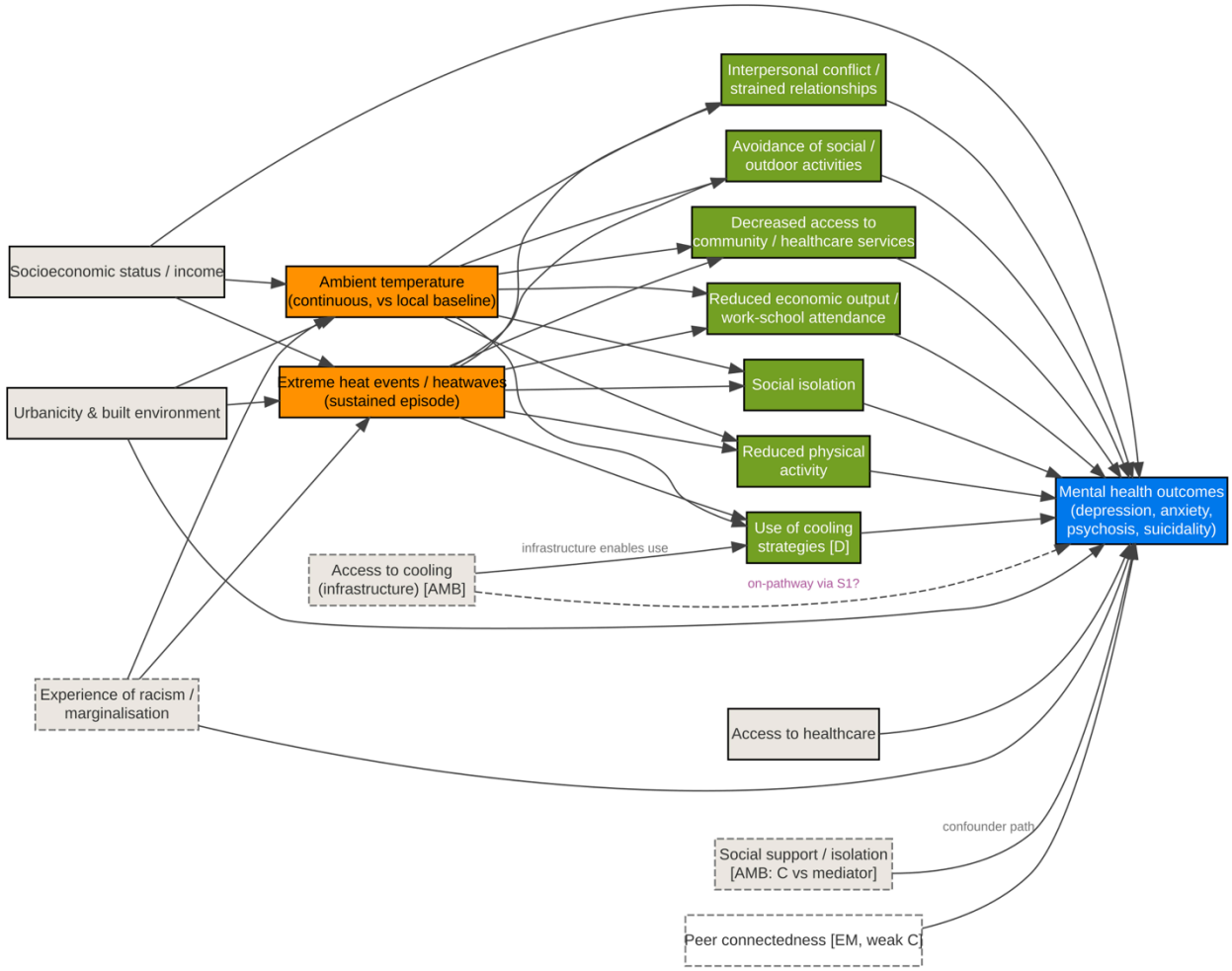


Figure 8: Social-behavioural mediator detail v1.1. The source distinguishes use of cooling from access to cooling infrastructure.

Assumptions illustrated in the main diagram

Source ID	What to look at in the diagram	Assumption or choice to discuss	Source status
A3	Two separate orange exposure nodes.	Ambient temperature and extreme heat should be treated as distinct exposure constructs.	Proposed.
A8	Dotted green links into psychological mediators.	Biological and social-behavioural pathways may influence psychological pathways.	Proposed; relationships need adjudication.
A9	Grey dotted line between depression and suicidality.	Represent the possible association without asserting a causal arrow.	Proposed; the line is a correlation annotation.

Planned evidence contributions

Planned contribution in the local materials	Proposed role	Issue to resolve in the ongoing review
Human observational studies	Heat–outcome associations; causal interpretation under stated identifying assumptions	Confounder roles, timing, exposure definition, and residual confounding.

Planned contribution in the local materials	Proposed role	Issue to resolve in the ongoing review
Quasi-experimental studies	Design-specific contrasts related to heat exposure	What the design identifies and whether its quantity matches the target question.
Cooling / heat-acclimatisation RCTs	Evidence about thermal interventions	Whether and how the intervention informs the heat-harm question; cooling without high heat exposure is not automatically its inverse.
Qualitative studies	Experiences, context, and candidate pathways	How these inform the proposed structure without being treated as effect estimates.
Animal experiments	Experimental heat effects on measured animal outcomes	Proxy validity, eligibility, and availability. The source package records no eligible studies in iteration 1; confirm the current position.

Detailed questions for follow-up — evaluation

These are candidate observations for a pilot, not additional questions for the initial meeting.

- Which assumptions or disagreements become more explicit, and do participants find that useful?
- Does the contribution map change how a stream is interpreted or which conclusion is considered warranted?
- How much preparation and meeting time is needed, and where is information duplicated?
- Can the record be maintained at the next update, with clear ownership and traceable revisions?
- What feedback do methodological, statistical, clinical, and lived-experience contributors give on its clarity?

Project materials and implementation details

The project materials include the completed protocol, draft DAG package v1.1, an assumption register, a provisional covariate-adjudication table, and a prefilled worksheet. The protocol records registration on 30 March 2026 (PROSPERO 1325620).

Evidence availability in those records is a dated snapshot. Current SET availability, the adoption date, meeting plans, and register ownership need confirmation with the review team. No effect estimates or meeting judgements are entered here.

The protocol makes convening a triangulation meeting conditional on sufficient evidence from at least two sources for an outcome. The register and evidence-gap record can be prepared before that condition is met.

The figures and working assumptions follow the draft LSR8 DAG package v1.1. Application status and provenance. The Briefing page holds the meeting questions.

About & references

About this draft

This v0.1 methodological discussion draft presents a causal-assumption framework linking a directed acyclic graph (DAG), an assumption register, and a **Summary of Evidence table (SET)**. The templates, terminology, and reporting examples are proposals for review, not validated guidance or adopted standards. The Briefing sets out the four feedback questions.

Methodological context

Turner RM, Spiegelhalter DJ, Smith GCS, Thompson SG. **Bias modelling in evidence synthesis.** *Journal of the Royal Statistical Society: Series A*. 2009;172(1):21–47. Read the paper.

The proposed framework is a qualitative counterpart to quantitative bias modelling: it makes causal structure, stream-specific contributions, connecting assumptions, and possible shared biases explicit. It does not estimate bias parameters or replace quantitative adjustment.

Estimand layer

The estimand specification and alignment guidance draws on:

- ICH E9(R1). **Addendum on Estimands and Sensitivity Analysis in Clinical Trials.** 2019. Guideline.
- Kahan BC, Hindley J, Edwards M, Cro S, Morris TP. **The estimands framework: a primer on the ICH E9(R1) addendum.** *BMJ*. 2024;384:e076316. Read the paper.
- Kahan BC, Devane D. **Estimands: what they are and why we should use them.** *Journal of Clinical Epidemiology*. 2026;189:112054. Read the paper.
- Keene ON, Lynggaard H, Englert S, Lanius V, Wright D. **Why estimands are needed to define treatment effects in clinical trials.** *BMC Medicine*. 2023;21:117. Read the paper.
- Kahan BC, Morris TP, White IR, Carpenter J, Cro S. **Estimands in published protocols of randomised trials: urgent improvement needed.** *Trials*. 2021;22:202. Read the paper.

The specification and alignment components are new methodological proposals developed alongside this framework; the full drafts are available as downloads and their development is recorded in the project's background documents.

Applications and evidence

TAAR1: retrospective illustration

The TAAR1 example reconstructs how the framework could have supported a completed review. Its diagram, assumption IDs, and statuses are illustrative retrospective judgements, not records of the original panel using these tools.

The worked SET reports estimates and 95% confidence intervals from the published triangulation and the living systematic review. It is not a current evidence update or an independent audit of those publications.

LSR8: ongoing application

For Heat & Mental Health, the framework is being introduced **after protocol completion, while the living systematic reviews are underway**. Proposed assumptions and priority adjudications remain open for review; they are not completed meeting conclusions.

Templates and figures

The protocol proposals reproduce proposed additions to the associations v3.2 and hypothesis-generating v2.2 templates. Extracts and blank working templates are available on the relevant proposal pages; the original internal Word drafts are not distributed here.

The TAAR1 DAG is a simplified reconstruction highlighting selected assumptions; its proxy connection is not a physical causal arrow. The six LSR8 diagrams reproduce the draft v1.1 figures without redrawing. The deliverables schematic illustrates the proposed workflow, not a causal model. Diagrams express assumptions to examine, not established effects.